

Physics 514, Problem Set 4: Curvature

due: *Tuesday, February 15, 5 PM*

Please place your solutions in the box outside my office (Rutherford 316). You are encouraged to discuss these problems with your colleagues, but you must write up your own solutions; the solutions you hand in should reflect your own work and understanding.

Reading: Chapter 3 of Carroll.

1. Chapter 3, Problem 1.
2. In this problem we explore symmetric spaces, which are the very simplest examples of curved spaces.

(a) Show that the quantity

$$g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}$$

has all of the symmetries of the Riemann tensor with respect to permutation of the indices, i.e. obeys equations (3.129)-(3.133) and (3.140) of the text.

(b) Show that if the Riemann curvature tensor obeys

$$R_{\mu\nu\rho\sigma} = C(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu})$$

then

$$C = \frac{R}{n(n-1)}$$

where n is the dimension of the manifold. Such spaces are called symmetric spaces.

- (c) Consider the sphere S^2 , with metric $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$. Compute the Christoffel symbols, Riemann tensor, Ricci tensor and Ricci scalar of S^2 and verify that it is a symmetric space. The sphere is the prototypical two dimensional symmetric space with positive curvature, $R > 0$. The other standard symmetric spaces are Euclidean space (with $ds^2 = dx^2 + dy^2$) which has $R = 0$ and hyperbolic space H_2 (with $ds^2 = dr^2 + \sinh^2 r d\theta^2$) which has $R < 0$. You do not need to verify this, though.
3. We are often used to thinking of curved spaces as embedded in higher dimensional space. Consider coordinates x^i on some m -dimensional space M with metric g_{ij} . We can think of an n -dimensional manifold N (with dimension $n < m$) as being embedded in M . If we

take coordinates y^a , $a = 1, \dots, n$ on N , then the embedding of N into M is defined by some functions $x^i(y^a)$. The metric on the submanifold N induced by this embedding is

$$ds^2 = g_{ij} \frac{\partial x^i}{\partial y^a} \frac{\partial x^j}{\partial y^b} dy^a dy^b$$

- (a) A one form v_i in M can be turned into a one form v_a on N by the equation

$$v_a = \frac{\partial x^i}{\partial y^a} v_i$$

v_a is known as the "pullback" of v_i . Show that v_a does indeed transform as a one-form on N .

- (b) A vector v^a in N can be turned into a vector v^i on M by the equation

$$v^i = \frac{\partial x^i}{\partial y^a} v^a$$

v^i is known as the "pushforward" of v^a . Show that v^i does indeed transform as a vector on M .

- (c) Consider flat three dimensional space, with metric δ_{ij} in Cartesian coordinates x^i . The unit sphere S^2 is defined by the equation $\delta_{ij} x^i x^j = 1$. Using the usual embedding

$$(x^1, x^2, x^3) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

verify that the equation above gives the usual metric on the sphere.

- (d) Consider three dimensional Minkowski space with metric $\eta_{\mu\nu}$, $\mu, \nu = 0, 1, 2$. We can define another space dS_2 as the solutions of the equation $\eta_{\mu\nu} x^\mu x^\nu = 1$. Show that

$$(x^0, x^1, x^2) = (\sinh t, \cosh t \cos \phi, \cosh t \sin \phi)$$

is a good parameterization of the points on dS_2 . What is the metric on dS_2 ? What is the signature of this metric? This space is known as two dimensional de Sitter space. There is very good evidence that our universe looked approximately like four dimensional de Sitter space during inflation, and pretty good evidence that it will look like de Sitter space again in a few hundred billion years.

- (e) Compute the Christoffel symbols, Riemann tensor, Ricci tensor and Ricci scalar of dS_2 and verify that it is a symmetric space.

4. Show that the metrics of the sphere S^2 and de Sitter space dS_2 are related by the complex coordinate transformation

$$\theta = it + \pi/2$$

This coordinate transformation changes the signature of the metric and is known as a Wick rotation. Note that this complex coordinate transformation is not physical; S^2 and dS_2 are very different spaces! Nevertheless, these coordinate transformations are sometimes useful for technical reasons. For example, this fact explains why many of your answers for problems 3(c) and 4(e) look so similar.

1. Carroll 3.1

We are given that:

$$\nabla_\sigma g_{\mu\nu} = 0. \quad (1)$$

Then we show:

$$\nabla_\sigma g^{\mu\nu} = 0. \quad (2)$$

Well, using (1):

$$\nabla_\sigma g_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} \nabla_\sigma g^{\alpha\beta} = 0$$

$$\Rightarrow \nabla_\sigma g^{\alpha\beta} = 0 \quad \text{as required.}$$

Next, we want to show:

$$\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0. \quad (3)$$

$$\text{Well } \epsilon_{\mu\nu\rho\sigma} \epsilon^{\mu\nu\rho\sigma} = -n!$$

$$\Rightarrow \nabla_\lambda (\epsilon_{\mu\nu\rho\sigma} \epsilon^{\mu\nu\rho\sigma}) = 0.$$

$$\Rightarrow \epsilon_{\mu\nu\rho\sigma} \nabla_\lambda \epsilon^{\mu\nu\rho\sigma} + \epsilon^{\mu\nu\rho\sigma} \nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0.$$

$$\text{But } \epsilon_{\mu\nu\rho\sigma} \nabla_\lambda \epsilon^{\mu\nu\rho\sigma} = \epsilon^{\mu\nu\rho\sigma} \nabla_\lambda \epsilon_{\mu\nu\rho\sigma}$$

$$\Rightarrow \epsilon^{\mu\nu\rho\sigma} \nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0.$$

Now let's go back.

We also know:

$$\nabla_\lambda (\epsilon_{\mu\nu\rho\sigma} \epsilon^{\alpha\beta\gamma\delta}) = 0$$

since it's some combination of δ -terms.

So:

$$\epsilon_{\alpha\beta\gamma\delta} \nabla_\lambda (\epsilon_{\mu\nu\rho\sigma} \epsilon^{\alpha\beta\gamma\delta}) = 0$$

$$\Rightarrow -n! \nabla_\lambda \epsilon_{\mu\nu\rho\sigma} + \epsilon_{\mu\nu\rho\sigma} (\epsilon_{\alpha\beta\gamma\delta} \nabla_\lambda \epsilon^{\alpha\beta\gamma\delta}) = 0.$$

But we know the second term is zero.

Thus: $\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0$ as required.

2. Symmetric spaces.

a. The relevant symmetries:

$$R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu} \quad (1)$$

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} \quad (2)$$

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma} \quad (3)$$

$$R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} = 0 \quad (4)$$

$$R_{\rho[\sigma\mu\nu]\tau} = 0 \quad (5)$$

$$\nabla_{[\tau} R_{\rho\sigma]\mu\nu} = 0. \quad (6)$$

Now consider:

$$T_{\rho\sigma\mu\nu} = g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}$$

We must show T satisfies (1) - (6). First:

$$T_{\rho\sigma\mu\nu} = g_{\rho\nu}g_{\sigma\mu} - g_{\rho\mu}g_{\sigma\nu}$$

$$= -T_{\rho\sigma\mu\nu} \quad \text{as required.}$$

(2) & (3) are similarly trivial. Moving on,

$$T_{\rho\sigma\mu\nu} + T_{\rho\mu\nu\sigma} + T_{\rho\nu\sigma\mu}$$

$$= \underbrace{g_{\rho\mu}g_{\sigma\nu}} - \underbrace{g_{\rho\nu}g_{\sigma\mu}} + \underbrace{g_{\rho\nu}g_{\mu\sigma}} - \underbrace{g_{\rho\sigma}g_{\mu\nu}} \\ + \underbrace{g_{\rho\sigma}g_{\nu\mu}} - \underbrace{g_{\rho\mu}g_{\nu\sigma}}$$

By virtue of the symmetry of the metric, the color-coded terms cancel out.

Thus:

$$T_{\rho\sigma\mu\nu} + T_{\rho\mu\nu\sigma} + T_{\rho\nu\sigma\mu} = 0.$$

Using the antisymmetry properties, however:

$$T_{\rho\sigma\mu\nu} + T_{\rho\mu\nu\sigma} + T_{\rho\nu\sigma\mu} \\ = \frac{1}{2} (T_{\rho\sigma\mu\nu} - T_{\rho\sigma\nu\mu} + T_{\rho\mu\nu\sigma} - T_{\rho\mu\sigma\nu} \\ + T_{\rho\nu\sigma\mu} - T_{\rho\nu\mu\sigma}) = 0.$$

So then $T_{\rho[\sigma\mu\nu]} = 0$ as required.

To prove (6), note that a covariant derivative ∇ of the metric is, by definition, zero. Since the tensor T is manifestly built by metric terms, it follows that:

$$\nabla_{[\alpha} T_{\rho\sigma]\mu\nu} = 0.$$

This completes the relevant symmetries.

b. Suppose:

$$R_{\mu\nu\rho\sigma} = C T_{\mu\nu\rho\sigma}$$

It follows by definition of the Ricci tensor:

$$R_{\nu\sigma} = C T^{\rho}_{\nu\rho\sigma}$$

$$= C (g^{\rho}_{\rho} g_{\nu\sigma} - g^{\rho}_{\sigma} g_{\nu\rho})$$

But $g^{\rho}_{\rho} = g^{\mu\rho} g_{\mu\rho} = \delta^{\mu}_{\mu} = n$. It follows:

$$R = g^{\nu\sigma} R_{\nu\sigma} = C (n g^{\nu\sigma} g_{\nu\sigma} - g^{\rho\nu} g_{\nu\rho})$$

$$= C(n^2 - n) = Cn(n-1)$$

And so we conclude:

$$C = \frac{R}{n(n-1)} \quad \text{as required.}$$

c. We are given:

$$ds^2 = d\theta^2 + \sin^2\theta d\phi^2$$

Whereas the Christoffel symbols are defined:

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2} g^{\rho\lambda} (\partial_{\mu} g_{\nu\lambda} + \partial_{\nu} g_{\lambda\mu} - \partial_{\lambda} g_{\mu\nu})$$

$$(\text{where } g_{\mu\nu} = \text{diag}(1, \sin^2\theta)).$$

A straightforward calculation gives:

$$\Gamma_{\nu\rho}^0 = \begin{pmatrix} 0 & 0 \\ 0 & -\sin\theta\cos\theta \end{pmatrix}$$

$$\Gamma_{\nu\rho}^1 = \begin{pmatrix} 0 & \cot\theta \\ \cot\theta & 0 \end{pmatrix}$$

The Riemann tensor is defined:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{\rho}_{\nu\sigma} - \partial_{\nu}\Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}$$

Then:

$$R^0{}_{0\mu\nu} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$R^0{}_{1\mu\nu} = \begin{pmatrix} 0 & \sin^2\theta \\ -\sin^2\theta & 0 \end{pmatrix}$$

$$R^1{}_{0\mu\nu} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$R^1{}_{1\mu\nu} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Now, by definition:

$$R_{\mu\nu} = R^0_{\mu 0 \nu} + R^1_{\mu 1 \nu}$$

So:

$$R_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}$$

And the Ricci scalar is:

$$R = g^{\mu\nu} R_{\mu\nu} = 2$$

S^2 is a symmetric space if:

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}g_{\sigma\nu} - g_{\mu\nu}g_{\sigma\rho}$$

where of course $R_{\mu\nu\rho\sigma} = g_{\alpha\mu} R^\alpha_{\nu\rho\sigma}$.

Because of symmetries it is already obvious equality holds for terms of the form $R_{\mu\rho\sigma}$ or otherwise $R_{\rho\sigma\mu}$.

The only other terms are all related to

R_{1010} by symmetry, & so it suffices to show equality holds just for this case.

In particular, $R_{1010} = \sin^2 \Theta$

On the other hand:

$$g_{11}g_{00} - g_{10}g_{01} = \sin^2 \Theta$$

And indeed, the results are in agreement.

Thus, S^2 is a symmetric space.

3. Submanifold Embeddings.

a. We define:

$$v_a = \frac{\partial x^i}{\partial y^a} v_i$$

So by the chain rule:

$$\begin{aligned} v_{a'} &= \frac{\partial x^i}{\partial y^{a'}} v_i \\ &= \frac{\partial y^a}{\partial y^{a'}} \left\{ \frac{\partial x^i}{\partial y^a} v_i \right\} \\ &= \frac{\partial y^a}{\partial y^{a'}} v_a \end{aligned}$$

which is indeed the one-form transformation.

b. Again via the chain rule:

$$v^{i'} = \frac{\partial x^{i'}}{\partial x^i} v^i$$

which is the right transformation for vectors.

c. The induced metric is:

$$ds^2 = g_{ij} \frac{\partial x^i}{\partial y^a} \frac{\partial x^j}{\partial y^b} dy^a dy^b$$

We define the embedding:

$$(x^1, x^2, x^3) = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

Then,

$$ds^2 = \frac{\partial x^1}{\partial y^a} \frac{\partial x^1}{\partial y^b} dy^a dy^b + \frac{\partial x^2}{\partial y^a} \frac{\partial x^2}{\partial y^b} dy^a dy^b$$

w/ the understanding that rigorously, this is two terms. But they cancel all the same.

$$+ \frac{\partial x^3}{\partial y^a} \frac{\partial x^3}{\partial y^b} dy^a dy^b$$

$$= \cos^2\theta \cos^2\phi (d\theta)^2 - \cancel{2\sin\theta \cos\theta \sin\phi \cos\phi d\theta d\phi}$$

$$+ \sin^2\theta \sin^2\phi (d\phi)^2 + \cos^2\theta \sin^2\phi (d\theta)^2$$

$$+ \cancel{2\sin\theta \cos\theta \sin\phi \cos\phi d\theta d\phi} + \sin^2\theta \cos^2\phi (d\phi)^2$$

$$+ \sin^2\theta (d\theta)^2$$

It's easy to see the color-coded terms combine to form:

$$ds^2 = (d\theta)^2 + \sin^2\theta (d\phi)^2 \quad \text{as expected.}$$

d. We must show two things:

1. All the points in the parametrization are in dS_2 .
2. All the points in dS_2 are in the parametrization for a unique choice of parameters.

Let's start with 2.

Suppose $\vec{x} = (x^0, x^1, x^2) \in dS_2$. Then:

$$-(x^0)^2 + (x^1)^2 + (x^2)^2 = 1$$

Then fix t s.t. $x^0 = \sinh t$, so that:

$$(x^1)^2 + (x^2)^2 = \cosh^2 t$$

This licenses us to fix φ s.t.:

$$x^1 = \cosh t \cos \varphi$$

which in turn forces:

$$x^2 = \cosh t \sin \varphi$$

They are manifestly unique & working back, it's easy to see 1. holds true.

ds^2 is calculated the same way, & we get:

$$ds^2 = -(dt)^2 + \cosh^2 t (dx)^2$$

Evidently it has Lorentzian signature.

e. By definition:

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} g^{\lambda\rho} (\partial_{\mu} g_{\nu\rho} + \partial_{\nu} g_{\rho\mu} - \partial_{\rho} g_{\mu\nu})$$

So:

$$\Gamma_{\mu\nu}^0 = \begin{pmatrix} 0 & 0 \\ 0 & \cosh t \sinh t \end{pmatrix}$$

$$\Gamma_{\mu\nu}^1 = \begin{pmatrix} 0 & \tanh t \\ \tanh t & 0 \end{pmatrix}$$

Next:

$$R^{\rho}_{\sigma\mu\nu} = \partial_{\mu} \Gamma_{\nu\sigma}^{\rho} - \partial_{\nu} \Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho} \Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \Gamma_{\mu\sigma}^{\lambda}$$

Then:

$$R^0_{0\mu\nu} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$R^0_{1\mu\nu} = \begin{pmatrix} 0 & \cosh^2 t \\ -\cosh^2 t & 0 \end{pmatrix}$$

$$R^1_{0\mu\nu} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$R^1_{1\mu\nu} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Next, $R_{\mu\nu} = R^0_{\mu 0 \nu} + R^1_{\mu 1 \nu}$

So:

$$R_{\mu\nu} = \begin{pmatrix} -1 & 0 \\ 0 & \cosh^2 t \end{pmatrix}$$

2. Finally:

$$R = g^{\mu\nu} R_{\mu\nu} = 2$$

Then, consider a non-vanishing component of the Riemann tensor:

$$R_{1010} = -\cosh^2 t$$

whereas:

$$g_{11}g_{00} - g_{10}g_{01} = -\cosh^2 t$$

Both tensors have one independent component, with the same symmetries. Thus:

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}g_{\sigma\nu} - g_{\mu\nu}g_{\sigma\rho}$$

& consequently dS_2 is maximally symmetric.

4. Wick Rotation

For S^2 we have:

$$ds^2 = (d\theta)^2 + \sin^2\theta (d\phi)^2$$

Letting $\theta = it + \pi/2$, $d\theta = i dt$. Further:

$$\sin(it + \pi/2) = \cos(it) = \cosh t$$

So:

$$ds^2 = -(dt)^2 + \cosh^2 t (d\phi)^2$$

which is indeed the metric of dS_2 .